

Chapter 3.

$$H^A \otimes H^B,$$

ONB $\{|e_a \otimes f_b\rangle \mid |e_a\rangle\}$: ONB of H^A , $\{|f_b\rangle\}$: ONB of H^B

$$|\Psi\rangle = \sum_{a,b} \Psi_{ab} |e_a \otimes f_b\rangle$$

inner product $\langle \Psi | \Phi \rangle = \sum_{a,b} \overline{\Psi_{ab}} \Phi_{ab}.$

→ positive definite

$$(\because |\Psi\rangle = \sum_{a,b} \Psi_{ab} |e_a \otimes f_b\rangle$$

$$\Rightarrow \langle \Psi | \Psi \rangle = \sum_{a,b} |\Psi_{ab}|^2 \geq 0$$

$$\langle \Psi | \Psi \rangle = 0 \quad \text{iff} \quad \forall \Psi_{ab} = 0 \quad \text{iff} \quad \Psi = 0$$

→ $\times$ affected by the choice of ONB.

(Fact: $|e_a\rangle = U^A |e_a\rangle$: ONB of Hilbert sp $\Leftrightarrow U$: Unitary operator)

Induced norm $\|\Psi\| = \sqrt{\langle \Psi | \Psi \rangle} = \sqrt{\sum_{a,b} |\Psi_{ab}|^2}.$

$$\rightarrow \|\varphi \otimes \Psi\| = \sqrt{\langle \varphi \otimes \Psi | \varphi \otimes \Psi \rangle} = \sqrt{\langle \varphi | \varphi \rangle \langle \Psi | \Psi \rangle} = \|\varphi\| \|\Psi\|$$

for $|\varphi\rangle \in H^A$, $|\Psi\rangle \in H^B$.

Summary

$H^A \otimes H^B$: Hilbert sp w. "tensor product of H^A & H^B "

$$\left\{ \begin{array}{l} \text{inner prod} \quad \langle \Psi | \Phi \rangle = \sum_{a,b} \overline{\Psi_{ab}} \Phi_{ab} \\ \text{norm} \quad \|\Psi\| = \sqrt{\sum_{a,b} |\Psi_{ab}|^2} \\ \text{ONB} \quad \{|e_a \otimes f_b\rangle \mid |e_a\rangle: \text{ONB of } H^A, |f_b\rangle: \text{ONB of } H^B\} \\ \dim(H^A \otimes H^B) = \dim H^A \cdot \dim H^B \quad (\text{for finite dim } H^A, H^B) \end{array} \right.$$

cf. $\{|e_a\rangle\}$: basis of H^A

$\{|f_b\rangle\}$: " H^B

$\{|g_c\rangle\}$: " H^C

$$\begin{aligned} \Rightarrow (H^A \otimes H^B) \otimes H^C &= H^A \otimes (H^B \otimes H^C) \\ &= H^A \otimes H^B \otimes H^C, \end{aligned}$$

$$\langle e_1 \otimes \psi_1 \otimes \chi_1 | e_2 \otimes \psi_2 \otimes \chi_2 \rangle = \langle e_1 | e_2 \rangle \langle \psi_1 | \psi_2 \rangle \langle \chi_1 | \chi_2 \rangle$$

$$|\Psi\rangle \in H^A \otimes H^B \otimes H^C \rightarrow |\Psi\rangle = \sum_{a,b,c} \Psi_{abc} |e_a \otimes f_b \otimes g_c\rangle$$

$$H^A \simeq \mathbb{C}^{n_A}$$

$$|e_i\rangle \mapsto \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

$$(H^A)^* \simeq H^A \simeq \mathbb{C}^{n_A}, \quad H^B \simeq \mathbb{C}^{n_B} \Rightarrow H^A \otimes H^B \simeq \mathbb{C}^{n_A n_B}$$

$$|e_i \otimes f_j\rangle = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}^{\leftarrow i} \quad |e_1 \otimes f_2\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \quad \dots, |e_a \otimes f_b\rangle = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}$$

$(a-1)n_B + b \rightarrow$

$$\langle e_a \otimes f_b | = (0 \cdots \underset{\substack{\uparrow \\ (a-1)n_B + b}}{1} \cdots 0).$$

$$\text{ex) } H^A = H^B \simeq \mathbb{C}^2$$

$$\text{ONB of } H^A, H^B : \begin{cases} |e_1\rangle = |f_1\rangle = |0\rangle, |e_2\rangle = |f_2\rangle = |1\rangle \end{cases} = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$$

$$\Rightarrow \text{ONB of } H^A \otimes H^B \simeq \mathbb{C}^4 : \begin{cases} |e_a \otimes f_b\rangle = |00\rangle, |01\rangle, |10\rangle, |11\rangle \end{cases}$$

$$= \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

$$\begin{aligned}
 \Rightarrow \quad | \psi \rangle &= a_1 |0\rangle + b_1 |1\rangle, & | \phi \rangle &= a_2 |0\rangle + b_2 |1\rangle, \\
 | \psi \otimes \phi \rangle &= (a_1 |0\rangle + b_1 |1\rangle) \otimes (a_2 |0\rangle + b_2 |1\rangle) \\
 &= a_1 a_2 |00\rangle + a_1 b_2 |01\rangle + b_1 a_2 |10\rangle + b_1 b_2 |11\rangle \\
 &= \begin{pmatrix} a_1 a_2 \\ a_1 b_2 \\ b_1 a_2 \\ b_1 b_2 \end{pmatrix}.
 \end{aligned}$$

$$\langle \psi \otimes \phi | = (a_1 a_2 \quad a_1 b_2 \quad b_1 a_2 \quad b_1 b_2)$$

Definition

- Tensor product of qubit spaces

$${}^1H^{\otimes n} := \underbrace{{}^1H \otimes \dots \otimes {}^1H}_{n \text{ times}} \quad \rightarrow \dim {}^1H^{\otimes n} : 2^n$$

- $(j+1)$ -th factor space 1H_j counts from the right!

$${}^1H^{\otimes n} = {}^1H_{n-1} \otimes {}^1H_{n-2} \otimes \dots \otimes \underbrace{{}^1H_j}_{(j+1)\text{th}} \otimes \dots \otimes {}^1H_0$$

* Basis of ${}^1H^{\otimes n}$

1. Computational basis.

For $x \in N_0$, $x < 2^n$,

$$x = x_{n-1} x_{n-2} \cdots x_1 x_0 \quad (2) \quad (x_i \in \{0, 1\})$$

$$\begin{aligned} \text{Define } |x\rangle &:= |x\rangle := |x_{n-1} x_{n-2} \cdots x_1 x_0\rangle := |x_{n-1}\rangle \otimes |x_{n-2}\rangle \otimes \cdots \\ &\quad \otimes |x_0\rangle. \\ &= \bigotimes_{j=n-1}^0 |x_j\rangle \end{aligned}$$

$$\rightarrow {}^1H^{\otimes n} = {}^1H_{n-1} \otimes \cdots \otimes {}^1H_j \otimes \cdots \otimes {}^1H_0$$

$$\Rightarrow |x_{n-1}\rangle \otimes \cdots \otimes |x_j\rangle \otimes \cdots \otimes |x_0\rangle$$

Lemma 3.7

$$\{ |x\rangle \in {}^1H^{\otimes n} \mid x \in N_0, x < 2^n \} : \text{ONB of } {}^1H^{\otimes n}$$

$$\begin{aligned} \text{Pf)} \quad 1) \quad \langle x | y \rangle &= \langle x_{n-1} x_{n-2} \cdots x_1 x_0 | y_{n-1} y_{n-2} \cdots y_1 y_0 \rangle \\ &= \langle x_{n-1} | y_{n-1} \rangle \langle x_{n-2} | y_{n-2} \rangle \cdots \langle x_0 | y_0 \rangle \\ &= \delta_{xy} \quad \therefore \text{orthonormal} \end{aligned}$$

$$2) \quad |\{ |x\rangle \}| = 2^n = \dim {}^1H^{\otimes n}.$$

$$8) \quad \ln^1 H \quad |0\rangle^1 = |0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle^1 = |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

$$\ln^1 H^{\otimes 2} \approx \mathbb{C}^4 \quad |0\rangle^2 = |00\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad |1\rangle^2 = |01\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix},$$

$$|2\rangle^2 = |10\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad |3\rangle^2 = |11\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

$$\ln^1 H^{\otimes 3} \approx \mathbb{C}^8 \quad |5\rangle^3 = |101\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\ln^1 H, \quad \text{let } \begin{cases} |e_1\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ |e_2\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix} \end{cases}$$

$$\Rightarrow |e_1 \otimes e_2\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix} \otimes \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix}$$

$$\langle e_1 \otimes e_2 | = \frac{1}{\sqrt{2}}(1 \ 1) \otimes \frac{1}{\sqrt{2}}(1 \ -1) = \frac{1}{2}(1 \ -1 \ 1 \ -1)$$

2. Bell basis : basis of $\mathbb{H}^{\otimes 2}$.

$$\left| \Phi^{\pm} \right\rangle := \frac{1}{\sqrt{2}} (|00\rangle \pm |11\rangle), \quad \left| \Psi^{\pm} \right\rangle := \frac{1}{\sqrt{2}} (|01\rangle \pm |10\rangle).$$

→ orthonormal

$$\langle \Phi^+ | \Phi^+ \rangle = \frac{1}{2} \langle 00+11 | 00+11 \rangle = \frac{1}{2} (1+0+0+1) = 1$$

$$\langle \Phi^+ | \Phi^- \rangle = \frac{1}{2} \langle \omega + 11 | \omega - 11 \rangle = \frac{1}{2} (1 + 0 + 0 - 1) = 0$$

rrr

Similarly $\langle \Phi^- | \Xi^- \rangle = \langle \Psi^\dagger | \Psi^\dagger \rangle = 1, \dots$

/ Computational B: sub-sys. of each elt is pure state.

(Bell B : " X pure state.

Ex) 111^2 $\left(\begin{array}{l} \text{Cupet B : } |00\rangle |01\rangle |10\rangle |11\rangle \\ \text{Bell B : } |\Phi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) \end{array} \right)$

3.3. States & Observables of composite systems.

Postulate 7

The Hilbert space of a composite sys. that consists of

$$\mathcal{H}^A \text{ and } \mathcal{H}^B \text{ is } \mathcal{H}^A \otimes \mathcal{H}^B$$

$\otimes \mathcal{H}^C$ $\otimes \mathcal{H}^D$

1. State

By Postulate 5, it is described by ρ on $\mathcal{H}^A \otimes \mathcal{H}^B$.

$$\rho = \sum_{j \in I} p_j |\psi_j\rangle\langle\psi_j|, \text{ where } \{|\psi_j\rangle | j \in I\} : \text{ONB of } \mathcal{H}^A \otimes \mathcal{H}^B,$$
$$p_j \in [0, 1], \sum_I p_j = 1.$$

* Convention: associate person - subsys.

Alice controls subsystem A
Bob controls subsystem B

2. Observables

let $\begin{cases} M^A: \text{operator on } \mathcal{H}^A \\ M^B: \text{operator on } \mathcal{H}^B \end{cases}$

$$\Rightarrow \text{Construct } \underline{M^A \otimes M^B} : \mathcal{H}^A \otimes \mathcal{H}^B \longrightarrow \mathcal{H}^A \otimes \mathcal{H}^B.$$
$$|e \otimes \psi\rangle \longmapsto M^A(|e\rangle) \otimes M^B(|\psi\rangle)$$
$$\begin{aligned} |\psi\rangle &\longmapsto \sum_{a,b} \psi_{ab} M^A(|e_a\rangle) \otimes M^B(|f_b\rangle) \\ \sum \psi_{ab} |e_a \otimes f_b\rangle &\end{aligned}$$

$\rightarrow M^A, M^B : \text{self-adj.} \Rightarrow M^A \otimes M^B : \text{self-adj.}$

($\because$ For any $|\varphi_1\rangle, |\varphi_2\rangle \in H^A, |\psi_1\rangle, |\psi_2\rangle \in H^B$,

$$\langle (M^A \otimes M^B)^* \varphi_1 \otimes \psi_1 | \varphi_2 \otimes \psi_2 \rangle$$

$$= \langle \varphi_1 \otimes \psi_1 | (M^A \otimes M^B) (\varphi_2 \otimes \psi_2) \rangle$$

$$= \langle \varphi_1 \otimes \psi_1 | M^A(\varphi_2) \otimes M^B(\psi_2) \rangle$$

$$= \langle \varphi_1 | M^A(\varphi_2) \rangle \langle \psi_1 | M^B(\psi_2) \rangle$$

$$= \langle (M^A)^* \varphi_1 | \varphi_2 \rangle \langle (M^B)^* \psi_1 | \psi_2 \rangle$$

$$= \langle (M^A)^* \varphi_1 \otimes (M^B)^* \psi_1 | \varphi_2 \otimes \psi_2 \rangle$$

$$= \langle M^A \varphi_1 \otimes M^B \psi_1 | \varphi_2 \otimes \psi_2 \rangle$$

$$= \langle (M^A \otimes M^B) (\varphi_1 \otimes \psi_1) | \varphi_2 \otimes \psi_2 \rangle$$

$\rightarrow$ matrix repn of $M^A \otimes M^B$?

suppose for $\{|\varphi_i\rangle\}, \{|\psi_j\rangle\}$: basis of H^A, H^B resp. w. dim n_A, n_B resp.

$$M^A = \begin{pmatrix} M^A_{11} & & \\ & \ddots & \\ & & M^A_{n_A n_A} \end{pmatrix}, \quad M^B = \begin{pmatrix} M^B_{11} & & \\ & \ddots & \\ & & M^B_{n_B n_B} \end{pmatrix}$$

Then w. basis $\{|a \otimes b\rangle\}$ of $H^A \otimes H^B$, ($\dim = n = n_A n_B$)

$$\begin{aligned}
 M^A \otimes M^B &= \sum_{a,a'=1}^{n_A} \sum_{b,b'=1}^{n_B} |a \otimes b\rangle \langle a \otimes b| (M^A \otimes M^B) |a' \otimes b'\rangle \langle a' \otimes b'| \\
 &= \sum \sum |a \otimes b\rangle \langle a \otimes b| M^A |a'\rangle \langle a'| \otimes M^B |b'\rangle \langle b'| \\
 &= \sum \sum |a \otimes b\rangle \underbrace{\langle a| M^A |a'\rangle}_{M^A_{aa'}} \underbrace{\langle b| M^B |b'\rangle}_{M^B_{bb'}} \langle a' \otimes b'| \\
 &= \sum_{a,a'=1}^{n_A} \sum_{b,b'=1}^{n_B} \underbrace{M^A_{aa'} M^B_{bb'}}_{(\alpha'-1)n_B + b'} |a \otimes b\rangle \langle a' \otimes b'|
 \end{aligned}$$

$$(\alpha'-1)n_B + b' \left(\begin{array}{cc|cc} \oplus & & & \oplus \\ & & M^A_{aa'} M^B_{bb'} & \\ \oplus & & & \oplus \end{array} \right)$$

$$M^A \otimes M^B = \quad (3.34)$$

$$\begin{array}{c}
 1 \\
 \vdots \\
 n_B \\
 n_B + 1 \\
 \vdots \\
 2n_B \\
 \vdots \\
 n_A n_B
 \end{array}
 \left(\begin{array}{ccc|ccc|ccc}
 1 & \dots & n_B & n_B + 1 & \dots & 2n_B & \dots & n_A n_B \\
 M^A_{11} M^B_{11} & \dots & M^A_{11} M^B_{n_B} & M^A_{12} M^B_{11} & \dots & M^A_{12} M^B_{n_B} & \dots & M^A_{1n_A} M^B_{1n_B} \\
 \vdots & & \vdots & \vdots & & \vdots & & \vdots \\
 M^A_{11} M^B_{n_B+1} & \dots & M^A_{11} M^B_{n_B n_B} & M^A_{12} M^B_{n_B+1} & \dots & M^A_{12} M^B_{n_B n_B} & \dots & M^A_{1n_A} M^B_{n_B n_B} \\
 \hline
 M^A_{21} M^B_{11} & \dots & M^A_{21} M^B_{n_B} & M^A_{22} M^B_{11} & \dots & M^A_{22} M^B_{n_B} & \dots & M^A_{2n_A} M^B_{1n_B} \\
 \vdots & & \vdots & \vdots & & \vdots & & \vdots \\
 M^A_{21} M^B_{n_B+1} & \dots & M^A_{21} M^B_{n_B n_B} & M^A_{22} M^B_{n_B+1} & \dots & M^A_{22} M^B_{n_B n_B} & \dots & M^A_{2n_A} M^B_{n_B n_B} \\
 \hline
 \vdots & & \vdots & \vdots & & \vdots & & \vdots \\
 M^A_{n_A 1} M^B_{11} & \dots & M^A_{n_A 1} M^B_{n_B n_B} & M^A_{n_A 2} M^B_{n_B+1} & \dots & M^A_{n_A 2} M^B_{n_B n_B} & \dots & M^A_{n_A n_A} M^B_{n_B n_B}
 \end{array} \right)$$

$$= \left(\begin{array}{c|c|c} M_{ii}^A & M^B & \dots & M_{in}^A & M^B \\ \hline & & & & \\ \hline M_{ni}^A & M^B & & M_{nn}^A & M^B \end{array} \right)$$

lem 3.13

$$\text{let } |\varphi_i\rangle, |\varphi_2\rangle \in H^A, |\psi_i\rangle, |\psi_2\rangle \in H^B$$

$$\Rightarrow |\varphi_i \otimes \psi_i\rangle \langle \varphi_2 \otimes \psi_2| = |\varphi_i\rangle \langle \varphi_2| \otimes |\psi_i\rangle \langle \psi_2|$$

pf) For any $|a_1\rangle, |a_2\rangle \in H^A, |b_1\rangle, |b_2\rangle \in H^B,$

$$\langle a_1 \otimes b_1 | \left(|\varphi_i \otimes \psi_i\rangle \langle \psi_2 \otimes \psi_2| \right) | a_2 \otimes b_2 \rangle$$

$$= \langle a_1 | \varphi_i \rangle \langle b_1 | \psi_i \rangle \langle \psi_2 | a_2 \rangle \langle \psi_2 | b_2 \rangle$$

$$\langle a_1 \otimes b_1 | \left(|\varphi_i \rangle \langle \varphi_2| \otimes |\psi_i \rangle \langle \psi_2| \right) | a_2 \otimes b_2 \rangle.$$

$$= \langle a_1 \otimes b_1 | \left(\underbrace{|\varphi_i \rangle \langle \varphi_2|}_{a_2} \otimes \underbrace{|\psi_i \rangle \langle \psi_2|}_{b_2} \right) | a_2 \otimes b_2 \rangle$$

$$= \langle a_1 | \varphi_i \rangle \langle b_1 | \psi_i \rangle \langle \varphi_2 | a_2 \rangle \langle \psi_2 | b_2 \rangle.$$

$$e_x) \quad {}^T H^{AB} = {}^T H \otimes {}^T H.$$

$$M^A \otimes M^B = \sigma_z^A \otimes \sigma_z^B \quad \left(\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right)$$

Consider Bell basis $\{ |\Phi^\pm\rangle = \frac{1}{\sqrt{2}} (|00\rangle \pm |11\rangle),$
 $|\Psi^\pm\rangle = \frac{1}{\sqrt{2}} (|01\rangle \pm |10\rangle) \}.$

$$\begin{aligned} \Rightarrow (\sigma_z^A \otimes \sigma_z^B) |\Phi^\pm\rangle &= \frac{1}{\sqrt{2}} \{ (\sigma_z \otimes \sigma_z) |00\rangle \pm (\sigma_z \otimes \sigma_z) |11\rangle \} \\ &= \frac{1}{\sqrt{2}} \{ \sigma_z |0\rangle \otimes \sigma_z |0\rangle \pm \sigma_z |1\rangle \otimes \sigma_z |1\rangle \} \\ &= \frac{1}{\sqrt{2}} \{ |0\rangle \otimes |0\rangle \pm (-|1\rangle) \otimes (-|1\rangle) \} \\ &= \frac{1}{\sqrt{2}} \{ |00\rangle \pm |11\rangle \} \\ &= |\Phi^\pm\rangle. \end{aligned}$$

$$(\sigma_z \otimes \sigma_z) |\Psi^\pm\rangle = -|\Psi^\pm\rangle$$

$$(\sigma_x \otimes \sigma_x) |\Phi^\pm\rangle = \pm |\Phi^\pm\rangle$$

$$(\sigma_x \otimes \sigma_x) |\Psi^\pm\rangle = \pm |\Psi^\pm\rangle$$

For basis $\{ |\Phi^+\rangle, |\Phi^-\rangle, |\Psi^+\rangle, |\Psi^-\rangle \}.$

$$\sigma_z \otimes \sigma_z = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

$$\sigma_x \otimes \sigma_x = \begin{pmatrix} 1 & & & \\ & -1 & & \\ & & 1 & \\ & & & -1 \end{pmatrix}$$

$$[\sigma_z \otimes \sigma_z, \sigma_x \otimes \sigma_x] = 0.$$

Table 3.1 State determination via joint measurement of $\sigma_z \otimes \sigma_z$ and $\sigma_x \otimes \sigma_x$

Measured value of		State after measurement
$\sigma_z \otimes \sigma_z$	$\sigma_x \otimes \sigma_x$	
+1	+1	$ \Phi^+\rangle$
+1	-1	$ \Phi^-\rangle$
-1	+1	$ \Psi^+\rangle$
-1	-1	$ \Psi^-\rangle$

Computing out measurements on subsystem H^A

→ Measurement of $M^A \otimes \mathbb{1}^B$ of composite sys. $H^A \otimes H^B$

$$|\psi\rangle \in H^A \otimes H^B,$$

||

$$\sum \psi_{ab} |e_a \otimes f_b\rangle$$

$$\begin{aligned} \Rightarrow \langle M^A \otimes \mathbb{1}^B \rangle_\psi &= \langle \psi | M^A \otimes \mathbb{1}^B | \psi \rangle \\ &= \sum \psi_{ab} \langle \psi | \underbrace{(M^A \otimes \mathbb{1}^B)}_{M^A e_a \otimes f_b} | e_a \otimes f_b \rangle \\ &= \sum_{a,b} \sum_{a',b'} \overline{\psi_{a'b'}} \psi_{ab} \underbrace{\langle e_{a'} \otimes f_{b'} | M^A e_a \otimes f_b \rangle}_{\langle e_{a'} | M^A e_a \rangle \underbrace{\langle f_{b'} | f_b \rangle}_{\delta_{bb'}}} \\ &= \sum_{a,a',b} \overline{\psi_{a'b}} \psi_{ab} \langle e_{a'} | M^A e_a \rangle \end{aligned}$$

$$\text{Claim) } \langle M^A \rangle_{\rho^A(\psi)} = \langle M^A \otimes \mathbb{1}^B \rangle_\psi,$$

$$\rho^A(\psi) := \sum_{a,a',b} \overline{\psi_{a'b}} \psi_{ab} |e_{a'}\rangle \langle e_a|$$

Claim) $\rho^A(\psi)$ is density oper.

pf) 1. self-adj

$$(\rho^A(\Psi))^* = \rho^A(\Psi)$$

2. positiv

$$\begin{aligned} \langle \varphi | \rho^A(\Psi) | \varphi \rangle &= \sum_{a,a',b} \overline{\Psi_{a'b}} \Psi_{ab} \langle e|e_a \rangle \langle e_{a'}|e \rangle \\ &= \sum_b \left(\sum_a \Psi_{ab} \langle e|e_a \rangle \right) \left(\sum_{a'} \overline{\Psi_{a'b}} \langle e_{a'}|e \rangle \right) \\ &= \sum_b \left| \sum_a \Psi_{ab} \langle e|e_a \rangle \right|^2 \geq 0 \end{aligned}$$

3. $\text{tr} = 1$

$$\begin{aligned} \text{tr}(\rho^A(\Psi)) &= \sum_{a',a,b} \overline{\Psi_{a'b}} \Psi_{ab} \underbrace{\text{tr}(|e_a\rangle \langle e_{a'}|)}_{\substack{\text{tr} \langle e|e_a \rangle \langle e_{a'}|e \rangle \\ \text{tr} \delta_{ac} \delta_{a'c}}} \\ &= \sum_{a,b} \overline{\Psi_{ab}} \Psi_{ab} \\ &= \|\Psi\|^2 = 1. \end{aligned}$$

$$\begin{aligned} \text{Pf)} \quad \langle M^A \rangle_{\rho^A(\Psi)} &= \text{tr}(\rho^A(\Psi) M^A) \\ &= \sum_c \langle e_c | \rho^A(\Psi) M^A | e_c \rangle \\ &= \sum_{a,a',b,c} \overline{\Psi_{a'b}} \Psi_{ab} \underbrace{\langle e_c | e_a \rangle \langle e_{a'} | M^A | e_c \rangle}_{\delta_{ac}} \end{aligned}$$

$$= \sum_{a, \alpha, b} \overline{\psi_{a\alpha b}} \psi_{a\alpha b} \langle e_\alpha | M^A | e_\alpha \rangle$$

$$= \langle M^A \otimes \mathbb{1}^B \rangle_\Psi.$$

$$\rho = |\Psi\rangle\langle\Psi|.$$

$$\text{cf. } \langle M^B \rangle_{\rho^B(\Psi)} = \langle \mathbb{1}^B \otimes M^B \rangle_\Psi$$